# Graph Invariants & Regular Maps

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Ryan Lam

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# Acknowledgement


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  supervised by Dr David Hume.

  (with Wanying Huang, Samuel Kelly)

- University of Bristol, School of Mathematics

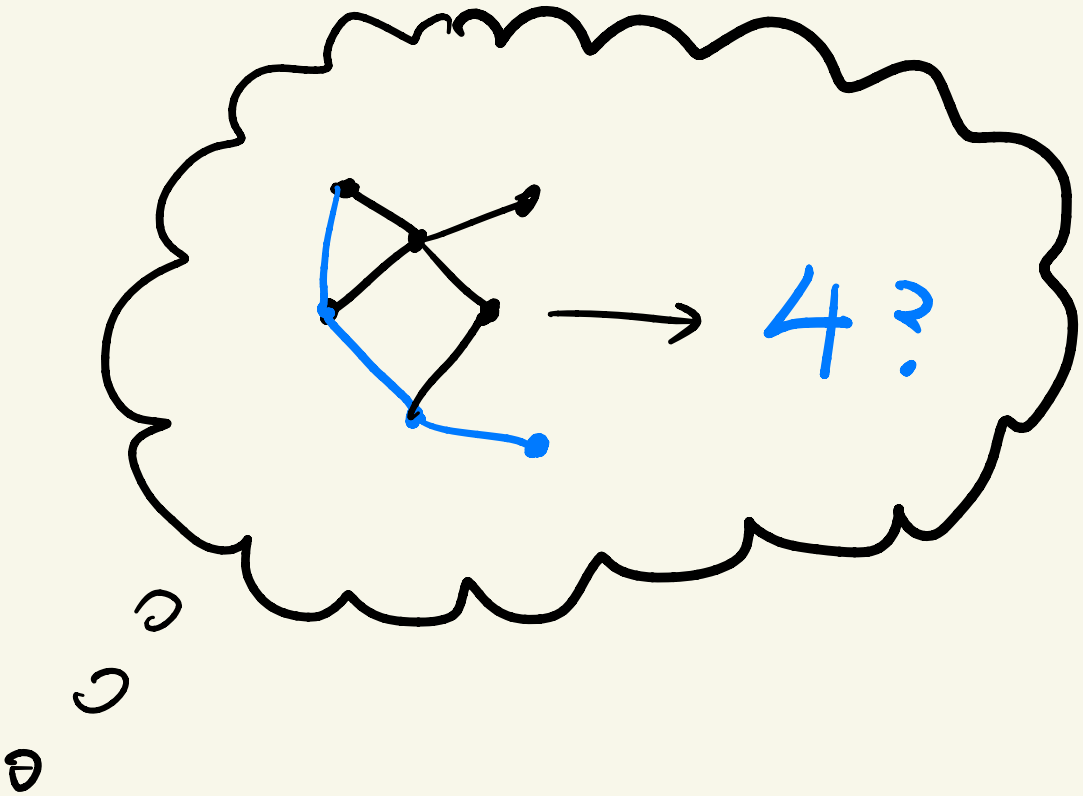


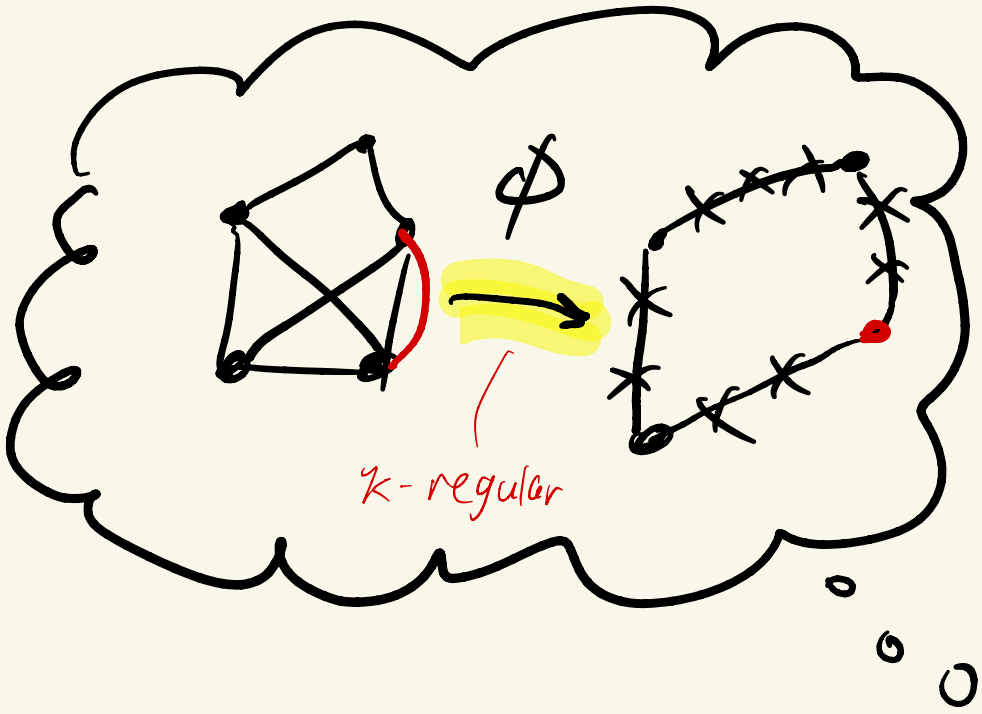

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# My Research

**Note:** All graphs are loop-less, undirected and connected.




## Graph Invariants

## "Nice" functions between graphs

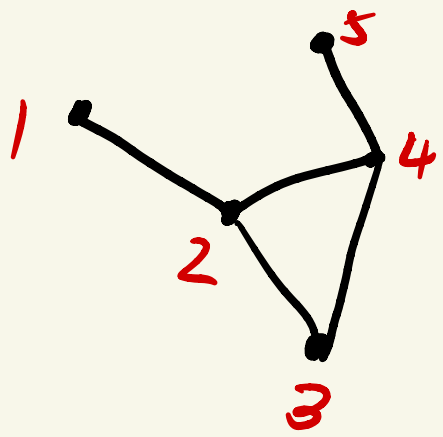
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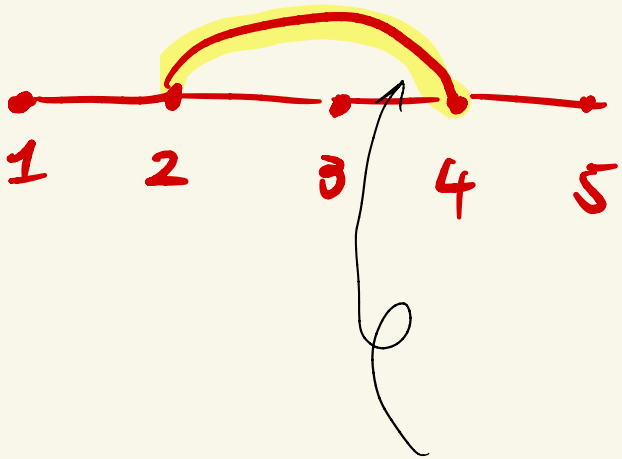
# I.

## Graph Invariants

( A function from [isomorphism classes] of graphs to a non-negative integer. )

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Infomally a "crossing"

Fixing a finite graph $G$, we can <u>number</u> it.

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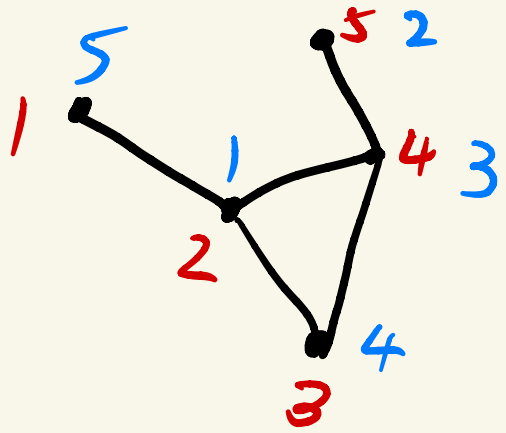
Interesting things to count:

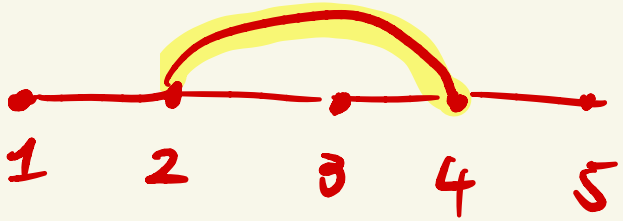
↳ Number of Crossings

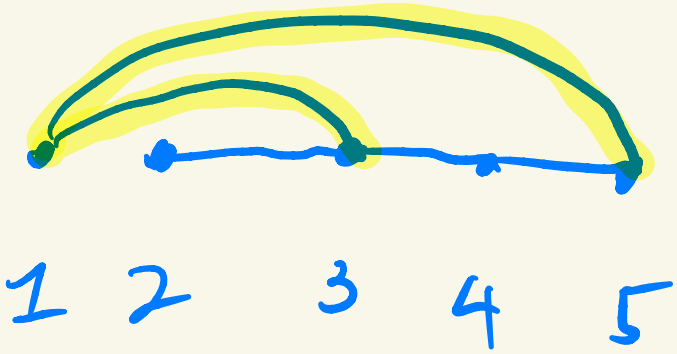
↳ Maximum width of Crossings

(and more)...

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However, there can be multiple numberings.

We take the minimum over all numberings.

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**Definition** The **cutwidth** of a **finite** graph $G$:

$$cw(G) := \min_f \max_{1 \leq k \leq |V_G|} \#\{(u,v) \in E_G \mid f(u) < k \leq f(v)\}$$

(Number of crossings)

The **bandwidth** of a **finite** graph $G$:

$$bw(G) := \min_f \max_{(v,w) \in E_G} |f(v) - f(w)|$$

(width of a crossing)

where $f$ ranges over all numberings.

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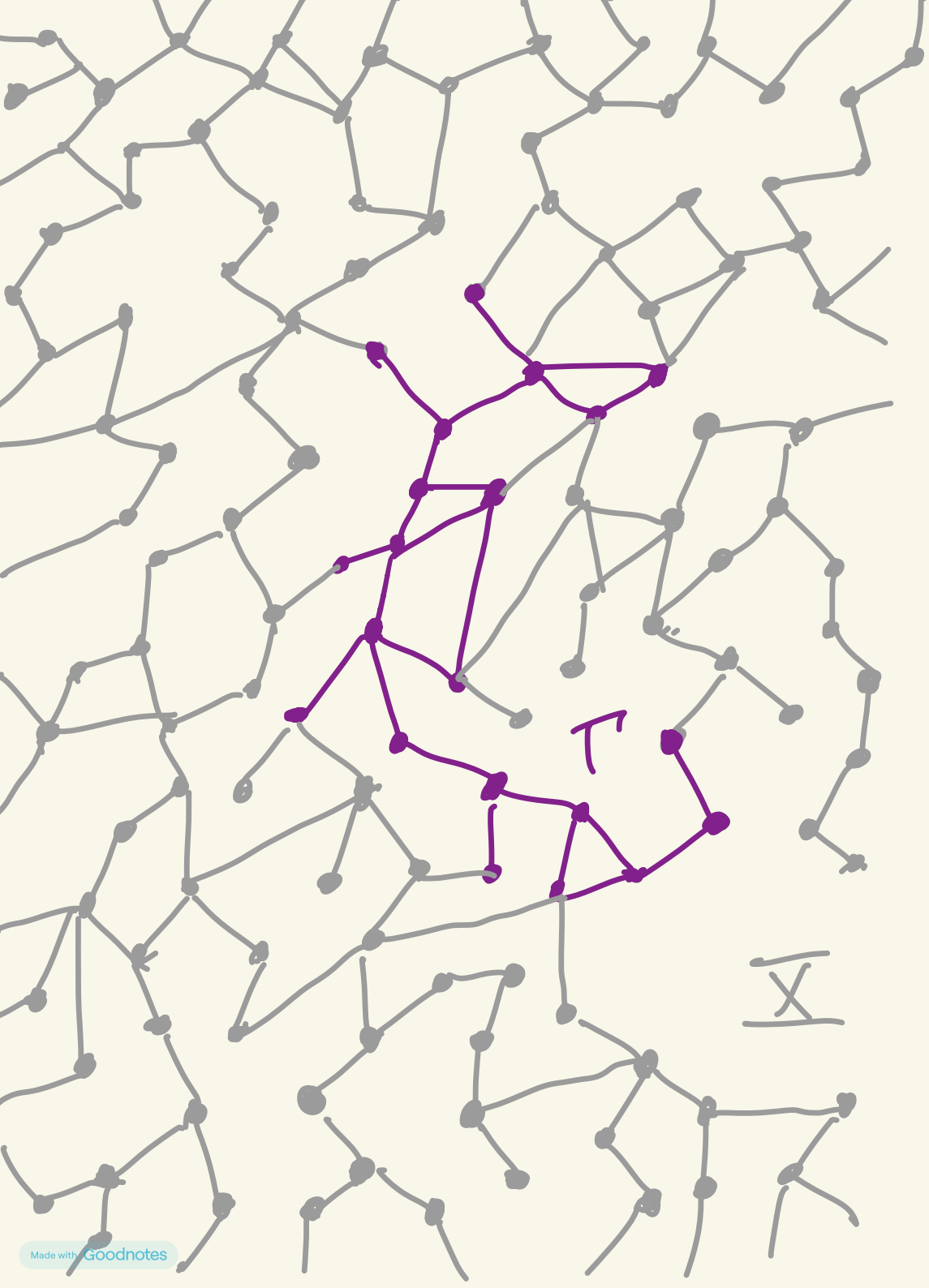
# Other Graph Invariants

- cut (G) - where it all started :

  [ Reference : Benjamini, Schramm, Timar, 2011 ]

- Search number $s(G)$
- Tree width, Pathwidth $tw(G)$, $pw(G)$
- Modified Cutwidth $mcw(G)$
- Topological Bandwidth $tbw(G)$

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# For Infinite Graphs

The $\mathcal{M}$-profile function $\mathbb{N} \to \mathbb{N}$ :

$$\mathcal{M}_X(n) := \max \{ \mathcal{M}(T) \mid T \ll X, \; |T| = n \}$$

(e.g cutwidth-profile, bandwidth profile... )

(Note: "cut-profile" is usually called separation profile $\mathrm{sep}_X$)

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Write $f \lesssim g$ if $f(n) = O(g(n))$ for all $n$:
($\exists$ constant $C$ s.t. $f(n) \le C g(n) + C$.)

**Theorem.** For any graph $G$:

$$sep_G \lesssim tw_G \lesssim cw_G \simeq s_G \simeq tbw_G \simeq pw_G \lesssim bw_G$$

Remark: Mostly collecting existing literature.

- Question: Is $sep \gtrsim tw$ ?

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# II.




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Any (connected) graph has a **geodesic metric** $d$.

**Definition.** $X, Y$ : Metric Spaces.

$f : X \to Y$ is **$\kappa$-regular** if

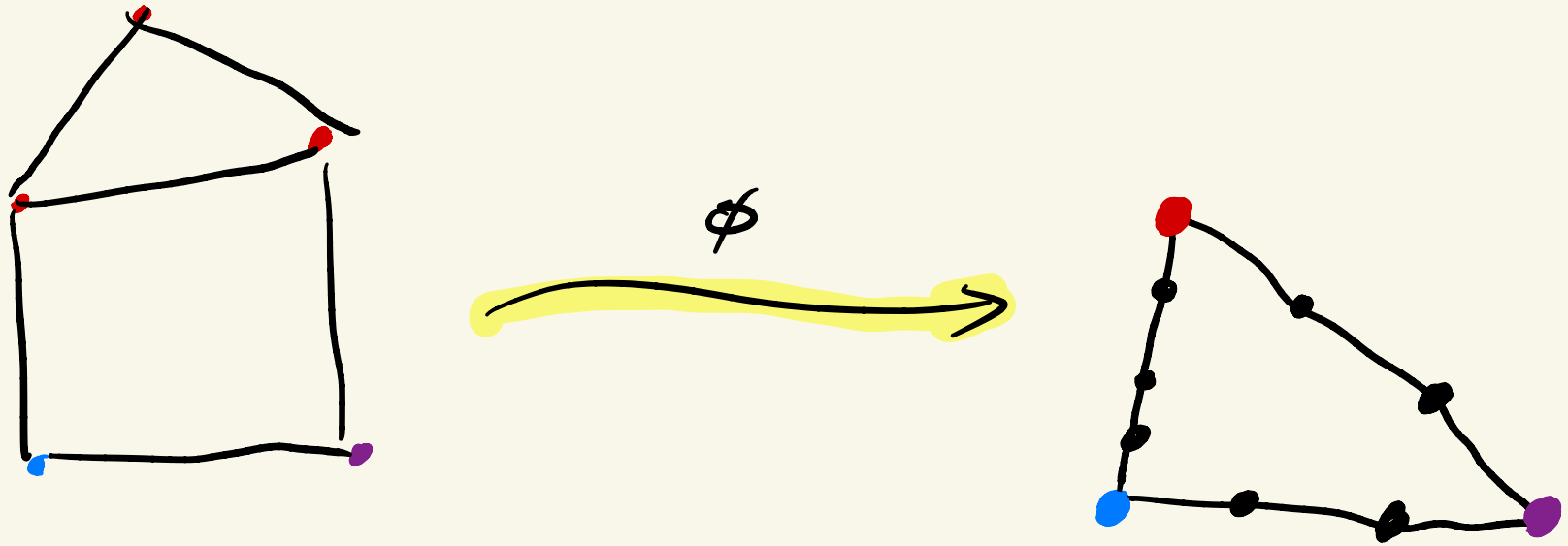
① $|f^{-1}(y)| \leq \kappa$ for all $y$, and

② $d_Y(f(x), f(y)) \leq (1+\kappa)\, d_X(x, y)$ for all $x, y$

A function is **regular** if it is $\kappa$-regular for some $\kappa$.

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# Intuition to regular maps.




Example of a 6-regular map.

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**Thm.** If there is a regular map of graphs $f : \overline{X} \to \overline{Y}$, then

$$\mu_{\overline{X}} \lesssim \mu_{\overline{Y}}$$

For $\mu \in \{\text{cut}, \text{cw}, \text{bw}, \text{s}, \text{pw}, \text{tw}, \text{tbw}\}$

**Remark.**

- $\mu = \text{cut}$ was resolved in [BST11].
- suffices to show for $\mu \in \{\text{bw}, \text{cw}, \text{tw}\}$.

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